Software Engineering & Design Principal Lab

Experiment 2 Template

Experiment: Identification of Problem, Objectives, Scope and Feasibility Analysis

Student/Team Details

Particular	Details
Experiment No.	2
Project Title	Disaster Management Information System (DMIS)
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Semester & Section	5th & D
Date of Submission	21 August 2026

1. Problem Identification

1.1 Background / Context

Domain: Disaster Management / Public Safety & Emergency Response.

India faces recurring natural and man-made disasters such as floods, earthquakes, cyclones, landslides, forest fires, and industrial accidents. Effective disaster management depends on the timely flow of information between citizens, local authorities, relief agencies (NDRF/SDRF, police, fire, medical), and government bodies at the district and state level. The proposed project is a web-based Disaster Management Information System (DMIS) that supports disaster reporting, resource coordination, alert dissemination, shelter/relief management, and post-disaster damage assessment.

1.2 Existing Problem

During a disaster, information about affected areas, victims needing help, availability of relief resources (food, medicines, shelters, rescue teams), and volunteer coordination is scattered across phone calls, radio, social media, and manual paperwork. This leads to delayed response, duplicated relief efforts in some areas while other areas remain unattended, difficulty in tracking real-time ground status, and lack of a single verified source of information for both citizens and disaster management authorities.

1.3 Existing Method / Current System

Currently, disaster response in most districts relies on manual coordination: control rooms receiving phone calls, WhatsApp/SMS-based alerts, physical registers for tracking relief camp inmates and supplies, and periodic situation reports compiled manually by field officers and sent up the chain via phone or email. Some information may also be shared informally through local news channels and social media, which is often unverified.

1.4 Limitations of the Existing System

1. No centralized, real-time database of affected areas, victims, and available resources.
2. Manual reporting causes delays between an incident occurring and authorities being informed.
3. Difficulty verifying rescue/relief requests, leading to duplication or missed cases.
4. No structured way for citizens to report incidents or request help digitally.
5. Relief resource allocation (shelters, food, medical aid) is not tracked systematically, causing shortages in some camps and surplus in others.

2. Problem Statement

Disaster management authorities, relief agencies, volunteers, and affected citizens currently lack a unified, real-time platform to report incidents, request and track relief resources, and disseminate verified alerts during natural or man-made disasters. The existing manual, phone-based, and fragmented communication methods result in delayed response times, inefficient allocation of relief resources, and lack of transparency in rescue and relief operations. A software-based Disaster Management Information System is required to centralize incident reporting, automate alert dissemination, coordinate relief resources in real time, and provide authorities with accurate, up-to-date situational data to make faster and better-informed decisions.

3. Objectives of the Project

S. No.	Objective
1.	To develop a centralized platform for reporting and tracking disaster incidents in real time.
2.	To automate the dissemination of disaster alerts and warnings to citizens in affected regions.
3.	To provide a mechanism for citizens to request rescue, medical, or relief assistance online.
4.	To enable authorities to manage and allocate relief resources (shelters, food, medical supplies) efficiently.
5.	To maintain a database of relief camps, volunteers, and rescue teams for coordinated response.
6.	To generate real-time situational reports and dashboards for disaster management authorities.
7.	To improve transparency and accountability in relief distribution and rescue operations.
8.	To reduce response time between incident reporting and relief action.

Objective Writing Tip

Action-oriented words used: To develop, To automate, To provide, To enable, To maintain, To generate, To improve, To reduce.

4. Scope of the Project

4.1 Target Users / Stakeholders

S. No.	User / Stakeholder	Role
1.	Citizen / Public User	Report incidents, request help, view alerts and shelter information
2.	Disaster Management Authority (Admin)	Verify reports, coordinate response, manage resources, publish alerts
3.	Volunteer / NGO Personnel	Register availability, get assigned to relief tasks, update ground status
4.	Rescue / Relief Team (NDRF, Police, Medical)	Receive assignments, update rescue/relief status in real time
5.	Relief Camp Manager	Manage shelter occupancy, inventory of food/medical supplies

4.2 Functional Scope

1. Citizen registration/login and incident reporting with location, description, and photos.
2. Real-time disaster alert and warning broadcast (SMS/email/push notification) to registered users in affected zones.
3. Request-for-help module allowing citizens to request rescue, medical aid, food, or shelter.
4. Admin dashboard to verify, prioritize, and assign incoming reports to relief/rescue teams.
5. Relief camp and resource management (shelter capacity, food stock, medical supplies).
6. Volunteer registration and task assignment module.
7. Reporting and analytics dashboard showing affected areas, resource status, and response statistics.

4.3 Data Scope

S. No.	Data / Information	Description
1.	User Data	Citizen, volunteer, and admin profiles with contact and role details
2.	Incident Reports	Type of disaster, location (GPS), description, photos, timestamp, status
3.	Relief Requests	Type of help needed, requester details, urgency level, assignment status
4.	Resource Inventory	Shelter capacity, food stock, medical supplies, vehicles, per relief camp
5.	Alerts & Notifications	Alert type, affected region, issue time, message content, broadcast log

4.4 Out of Scope

1. Real-time satellite/weather data integration for automated disaster prediction.
2. Drone-based or IoT sensor-based field data collection.
3. Integration with national emergency helpline numbers (e.g., 112) and government databases.
4. Payment gateway for donation collection and fund management.

Note: Features outside the scope may be mentioned as future enhancements.

5. Technical Feasibility Analysis

Technical feasibility determines whether the proposed system can be technically developed and implemented with the available resources.

5.1 Proposed Technology Stack

Component	Proposed Technology
Frontend	HTML5, CSS3, JavaScript, React.js / Bootstrap
Backend	Node.js with Express.js (or Java Spring Boot / Python Django as alternative)
Database	MySQL / MongoDB
Programming Language(s)	JavaScript, SQL
Development Tools/IDE	Visual Studio Code, Git & GitHub, Postman
Other Technologies	Google Maps API (location tracking), Twilio/Firebase (SMS & push alerts)

5.2 Hardware Requirements

Hardware	Requirement
Processor	Intel i3 (or equivalent) and above
RAM	4 GB minimum (8 GB recommended)
Storage	20 GB free disk space
Other Hardware	Stable internet connection; GPS-enabled device for mobile access

5.3 Software Requirements

Software	Version / Requirement
Operating System	Windows 10/11, Linux, or macOS
IDE / Editor	Visual Studio Code
Database	MySQL 8.0 / MongoDB 6.0
Framework / Libraries	Express.js, React.js, Bootstrap 5
Other Software	XAMPP/Node runtime, Postman, Git

5.4 Technical Skills Required

- Web development (HTML, CSS, JavaScript, React)
- Backend development and REST API design (Node.js/Express)
- Database design and management (SQL/NoSQL)
- Basic knowledge of cloud/notification services (Firebase, Twilio, Maps API)

Technical Feasibility Assessment

All required technologies (React, Node.js, MySQL/MongoDB) are open-source, well-documented, and already familiar to the team from coursework. The hardware and software requirements are minimal and available on standard college lab systems and personal laptops. Third-party APIs used (Maps, SMS/notification) offer free-tier access sufficient for an academic project. Hence, the project is technically achievable within the given time and resource constraints.

Conclusion:

☒ Technically Feasible ☐ Technically Not Feasible

Justification: The chosen technology stack is open-source, widely supported, and matches the skill set the team has already developed; no specialized or costly infrastructure is required.

6. Operational Feasibility Analysis

Operational feasibility determines whether the proposed system can be effectively used and integrated into the intended environment.

6.1 User Acceptance

Citizens are increasingly familiar with mobile/web apps for alerts and services (e.g., weather apps, government portals), and disaster authorities already use digital tools such as WhatsApp and spreadsheets, so a purpose-built platform is a natural and welcome upgrade. Given the direct benefit of faster help during emergencies, user acceptance is expected to be high.

6.2 Ease of Use

The system will feature a simple, minimal-step interface: a one-tap 'Report Incident' / 'Request Help' button for citizens, and a clear dashboard with filters and status indicators for administrators. Multilingual support and large, icon-based buttons will make it usable even for less tech-savvy citizens during stressful situations.

6.3 Existing Workflow

The system digitizes and streamlines the current manual, phone-based workflow rather than replacing human decision-making. Control room staff will continue to make the final call on resource deployment, but the platform will centralize incoming reports and resource data so decisions are faster and better informed.

6.4 Training Requirements

Citizens will require little to no training due to the simple interface. Admins, volunteers, and relief camp managers will need a short (1-2 hour) orientation session covering dashboard navigation, incident verification, and resource-update procedures.

6.5 Expected Operational Benefits

1. Faster incident reporting and quicker rescue/relief response times.
2. Reduced duplication of relief efforts and better resource utilization.
3. Improved transparency and traceability of relief distribution.
4. Better-informed decision-making through consolidated, real-time dashboards.

Operational Feasibility Assessment

☒ Operationally Feasible ☐ Operationally Not Feasible

Justification: The system complements, rather than disrupts, existing disaster response workflows, requires minimal training, and offers clear, tangible benefits that encourage adoption by both citizens and authorities.

7. Overall Feasibility Summary

Feasibility Area	Status	Justification
Technical Feasibility	Feasible	Open-source, well-documented tech stack matching team skills
Operational Feasibility	Feasible	Minimal training needed; complements existing workflow

Overall Conclusion

Based on the technical and operational feasibility analysis, the proposed project is feasible because it uses readily available, low-cost technology that the team is already skilled in, requires minimal

hardware/training, and directly addresses a real and urgent gap in current disaster response coordination.

8. Expected Outcome

The completed Disaster Management Information System is expected to provide a centralized, real-time platform where citizens can report disasters and request help, authorities can verify and coordinate relief operations, and relief resources (shelters, food, medical aid) are tracked and allocated efficiently. The system aims to reduce response time, minimize duplication of relief efforts, and improve overall transparency and coordination during disaster response.

9. Future Enhancements (Optional)

1. Integration with weather/seismic data APIs for early-warning prediction.
2. Mobile app version with offline-mode incident reporting.
3. AI-based severity classification and resource-allocation recommendation.